04. Root Finding: More Newton-Raphson

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Problem to solve: Given a $u \in [0,1]$, we want find a z s.t. $F(z) - u = 0$

i.e. Find the root.

One common tool: Newton-Raphson

- Start at some $z_0 \in \mathbb{R}$
- update $z_1 = z_0 - \dfrac{F(z_0) - u}{f(z_0)}$

CDF

$\left.\dfrac{d}{dz} F(z)\right|_{z=z_0}$

PDF

- Continue in this way until "convergence"

↳ in practice, pick a threshold τ "small #" and stop when $\underbrace{|z_i - z_{i-1}|}_{\text{change at the ith step is "small"}} < \tau$

* Note, we generate a $U \sim Unif[0,1]$ to start, then treat it as a constant w.r.t. z

Note: When NR works it works very well!

we get Quadratic Convergence

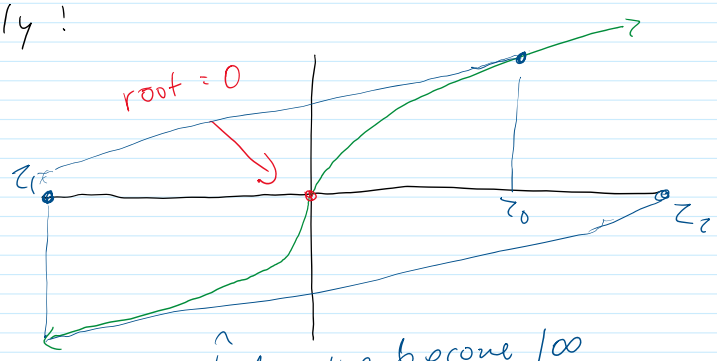
if z_∞ is the root then

$$\underbrace{|z_{i+1} - z_\infty|}_{\text{error from i+1st to the root}} = C_i \underbrace{(z_i - z_\infty)^2}_{\text{previous error}}$$

Note: Depending on F and z_0, this could fail very quickly!

Example 1: Square Root

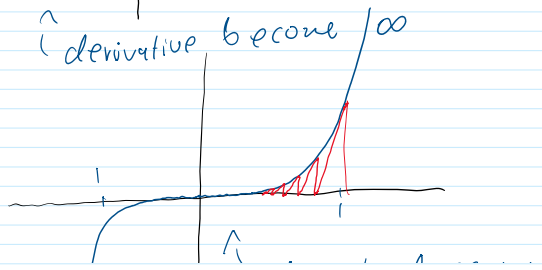
$$G(x) = \begin{cases} \sqrt{x}, & x \geq 0 \\ -\sqrt{-x} & x < 0 \end{cases}$$



- $\dfrac{d}{dx}\sqrt{x} = \left.\dfrac{1}{2\sqrt{x}}\right|_0 = \dfrac{1}{0}$

derivative become ∞

Example 1A: $G(x) = \begin{cases} x^{2k}, & x \geq 0 \\ -x^{2k} & x < 0 \end{cases}$



Example 1A: $G(x) = \begin{cases} x, & x > 0 \\ -x^{2k} & x < 0 \end{cases}$

$$x_i = x_{i-1} - \frac{G(x_{i-1})}{G'(x_{i-1})}$$

Extensions: There are many extensions to NR.

e.g. Halley's Method

$$x_{i+1} = x_i - \frac{2F(x_i)F'(x_i)}{2F'(x_i)^2 - F(x_i)F''(x_i)}$$

2nd derivative

General Rule: Nicer Functions and Good starting points will converge quickly.

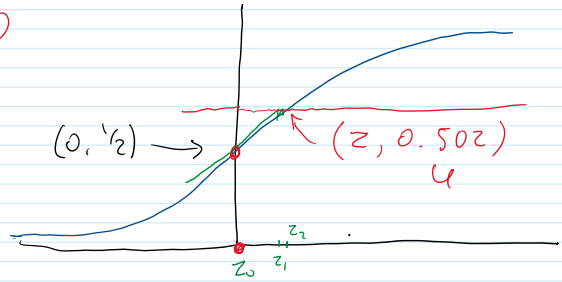
See "Householder Methods"

Pick a good starting Point!!!

```
# Pick a reasonable starting value of 0
rootFind.nr( 0, normal.cdf, normal.pdf, tol = 1e-4, max.it=100, PrintPath=T )

[1] "U = 0.502652446273714"
[1] "0 0.00664869682663191 FALSE"
[1] "0.00664869682663191 0.00664874581185144 TRUE"

[1] 0.006648746
```

Good Start at 0

Did not hit my threshold for convergence



$z_0 = 0$, $z_1 = 0.00665$

$z_2 = 0.00665$

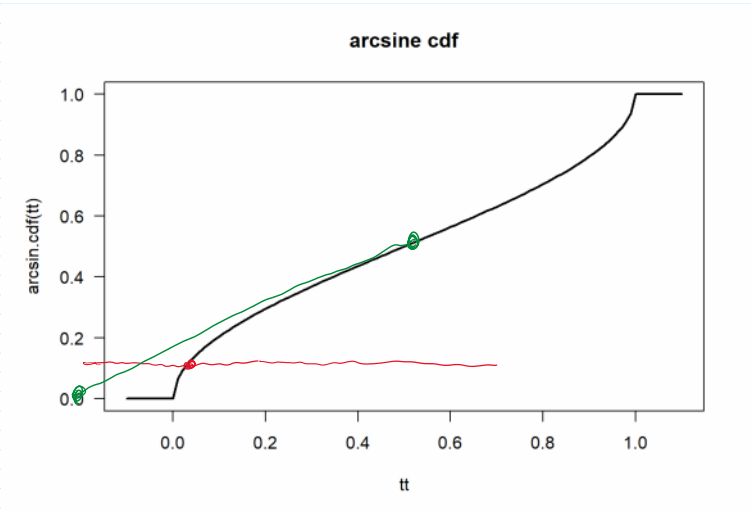
```
# Pick a bad starting value of 4
rootFind.nr( 4, normal.cdf, normal.pdf, tol = 1e-4, max.it=100, PrintPath=T )

[1] "U = 0.706531303469092"
[1] "4 -2188.60651778761 FALSE"
[1] "-2188.60651778761 Inf FALSE"

[1] Inf
```

Funct Derivative

Bad

$z_0 = 4$, $z_1 = -2188.6$

$z_2 = \infty$



NR does not know Boundaries!